(A) Data and analysis sets

Representation set: 278 cases
Centres 2–5: 49 / 41 / 71 / 117
E1, E3-site, PCA, LEACE

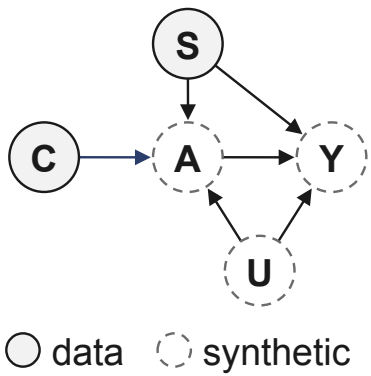
- **Primary causal set (W = 30): 158 cases**
  28 / 34 / 44 / 52
  - Single-file: 123 cases (sensitivity)
- W = 5 set: 201 cases (sensitivity)
- W = 0 set: 204 cases (sensitivity)

(B) Time structure and data-generating process

d(a) = δ·a, δ = 1.05

W pre-window steps	**K window steps** policy applied	P post-window steps	→ Y

1 step = 5 s of video
Window start w = s + O, O = −15, …, 15
s: step at which S switches 0 → 1



(C) Adjustment arms (each also adjusts for past actions H)

Arm	C	S	U
NV			
SITE	•		
ORACLE−C		•	
ORACLE	•	•	
FULL	•	•	•

Arm	R	R, site removed
FM-P · FM-S · REF	•	
FM-P−C · FM-S−C · REF−C		•
FM-P-nat	•	

(D) Workflow and registrations

records committed automatically

Protocol frozen 2026-10-05 → **Code lock** 2026-10-05 → **Steps 4–5** E1–E3 checks 2026-10-12 → **Step 6** causal grid to 2026-10-18 → **Step 7** verdicts 2026-10-19

Exploratory, outside the protocol (dashed); plans registered on OSF: X1/X2 during step 6, B after step 7

- **X1/X2 plan** 2026-10-13 osf.io/5c627
- **B plan** 2026-10-19 osf.io/g9ymw